

Date : Ex :	Create a Dashboard Using POWER BI
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AIM :

To develop an interactive Instagram Analytics Dashboard using Microsoft Power BI to analyze follower count, engagement rate, content category, media type, post performance, and engagement trends using appropriate charts, KPI cards, tables, and slicers.

ALGORITHM :

1. Open Microsoft Power BI Desktop.
2. Import the Instagram Analytics dataset into Power BI.
3. Examine the available fields such as **follower_count**, **engagement_rate**, **content_category**, **media_type**, **post_id**, and date fields.
4. Select suitable visualizations based on the type of analysis required.
5. Create a KPI Card to display the total follower count.
6. Create another KPI Card to display the total engagement rate.
7. Create a Line Chart to analyze engagement trends over time.
8. Create a Column Chart to compare engagement across content categories.
9. Create a Donut/Pie Chart to analyze engagement based on media type.
10. Create a Table to display post IDs and their engagement rates.
11. Create a List Slicer for **content_category**.
12. Create another List Slicer for **media_type**.
13. Arrange and format all visualizations on a single dashboard.
14. Test the slicers and verify that the visualizations respond interactively.
15. Analyze the completed dashboard and derive meaningful insights.

PROCEDURE :

Diagram 1: KPI Card – Total Followers

Procedure :

1. Open the Power BI report.
2. Select the Card visual from the Visualizations pane.
3. Drag the **follower_count** field from the Data pane.
4. Drop it into the Value field of the Card.
5. Power BI automatically performs the aggregation, such as Sum of follower_count.
6. Rename the card title as Total Followers.
7. Format the card for better visibility.

Output :

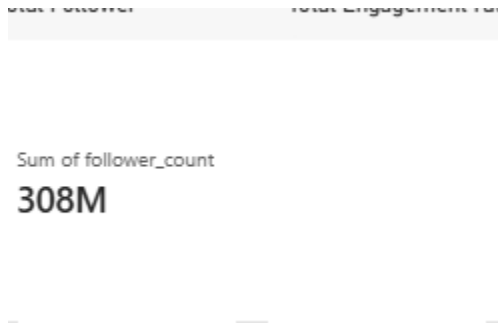


Diagram 2: KPI Card – Total Engagement

Procedure

1. Select a new Card visual.
2. Locate the **engagement_rate** field in the Data pane.
3. Drag **engagement_rate** into the Value field.
4. Verify that Power BI calculates the required aggregation.
5. Rename the card as Total Engagement Rate.
6. Format the card appropriately.

Output

The KPI card displays the calculated engagement value for the Instagram posts.

Total Engagement rate

Sum of engagement_rate

1.26K

Diagram 3: Line Chart – Engagement Over Time

Procedure

1. Select the Line Chart visualization.
2. Drag the appropriate date field, such as `post_date`, into the X-axis.
3. Drag `engagement_rate` into the Y-axis.
4. Power BI plots the engagement values against the corresponding dates.
5. Add a suitable title such as Engagement Over Time.
6. Format the chart for clear visualization.

Output

A line chart is generated showing the trend of Instagram engagement over time.

Sum of engagement_rate

by post_date

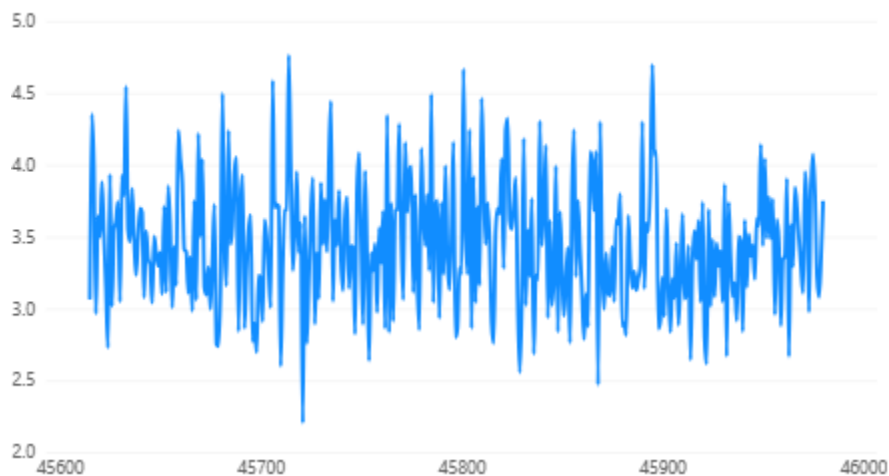


Diagram 4: Column Chart – Engagement by Content Category

Procedure

1. Select the Clustered Column Chart.
2. Drag `content_category` into the X-axis.
3. Drag `engagement_rate` into the Y-axis.
4. Power BI calculates the engagement for each content category.
5. Add the title Engagement by Content Category.
6. Format the chart for better readability.

Output

A column chart is generated to compare engagement among different content categories.

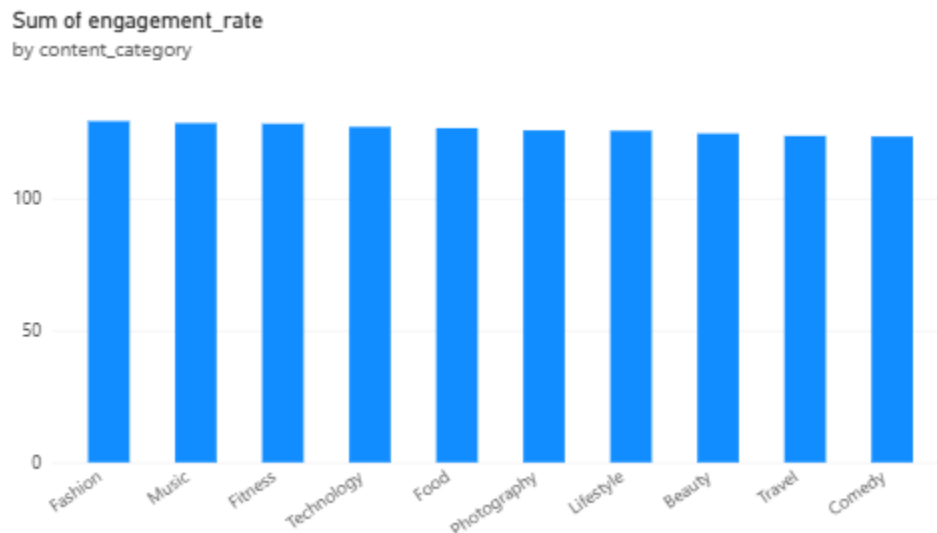


Diagram 5: Donut/Pie Chart – Engagement by Media Type

Procedure

1. Select the Donut Chart or Pie Chart visual.
2. Drag `media_type` into the Legend field.
3. Drag `engagement_rate` into the Values field.
4. Power BI creates separate sections for each media type.
5. Add the title Engagement by Media Type.
6. Format the chart and enable data labels if required.

Output

A donut/pie chart is generated showing the engagement distribution among different media types.

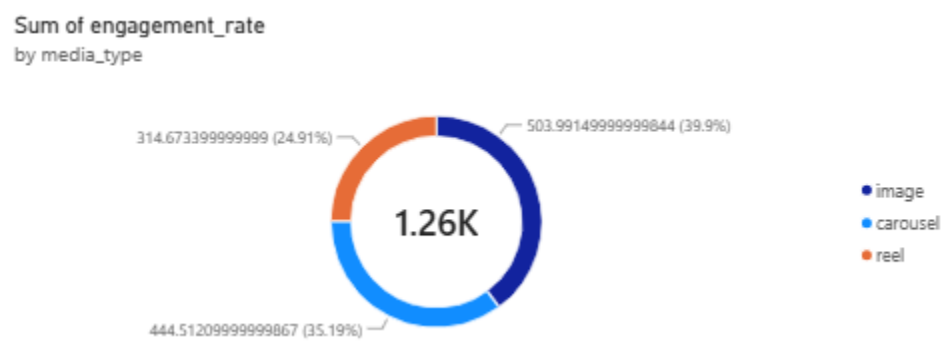


Diagram 6: Table – Top Posts by Engagement

Procedure

- 1. Select the Table visualization.
- 2. Drag `post_id` into the table.
- 3. Drag `engagement_rate` into the table.
- 4. Sort the `engagement_rate` column in descending order.
- 5. The posts with higher engagement appear at the top.
- 6. Rename the visual as Top Posts by Engagement.

Output

A table is generated displaying the post IDs and their corresponding engagement rates, allowing high-performing posts to be identified.

post_id	Sum of engagement_rate
IG0005227	0.27
IG0017615	0.27
IG0009304	0.25
IG0017637	0.25
IG0017987	0.25
IG0025791	0.25
IG0028356	0.24
IG0007780	0.24
IG0019641	0.24
IG0007979	0.24
IG0029802	0.24
Total	1,263.18

Diagram 7: Content Category Slicer

Procedure

1. Select the Slicer visual.
2. Choose List Slicer.
3. Drag **content_category** into the Field area.
4. The available content categories are displayed as a list.
5. Select any category to filter the dashboard.
6. Clear the selection to return to the complete dataset.

Output

An interactive Content Category slicer is created, allowing users to filter the dashboard based on content category.

content_category

- ☐ Beauty
- ☐ Comedy
- ☐ Fashion
- ☐ Fitness
- ☐ Food
- ☐ Lifestyle
- ☐ Music

Diagram 8: Media Type Slicer

Aim

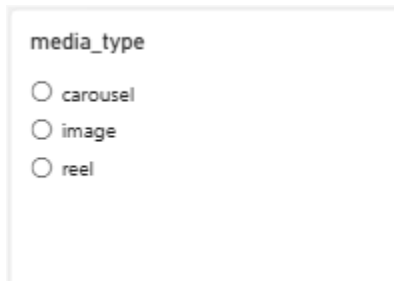
To provide an interactive filter for analyzing Instagram performance based on media type.

Procedure

1. Select another Slicer visual.
2. Choose List Slicer.
3. Drag **media_type** into the Field area.
4. The available media types are displayed.
5. Select a media type to filter the dashboard.
6. Observe the changes in the associated charts and KPI cards.

Output

An interactive Media Type slicer is created to filter the dashboard according to the selected media type.



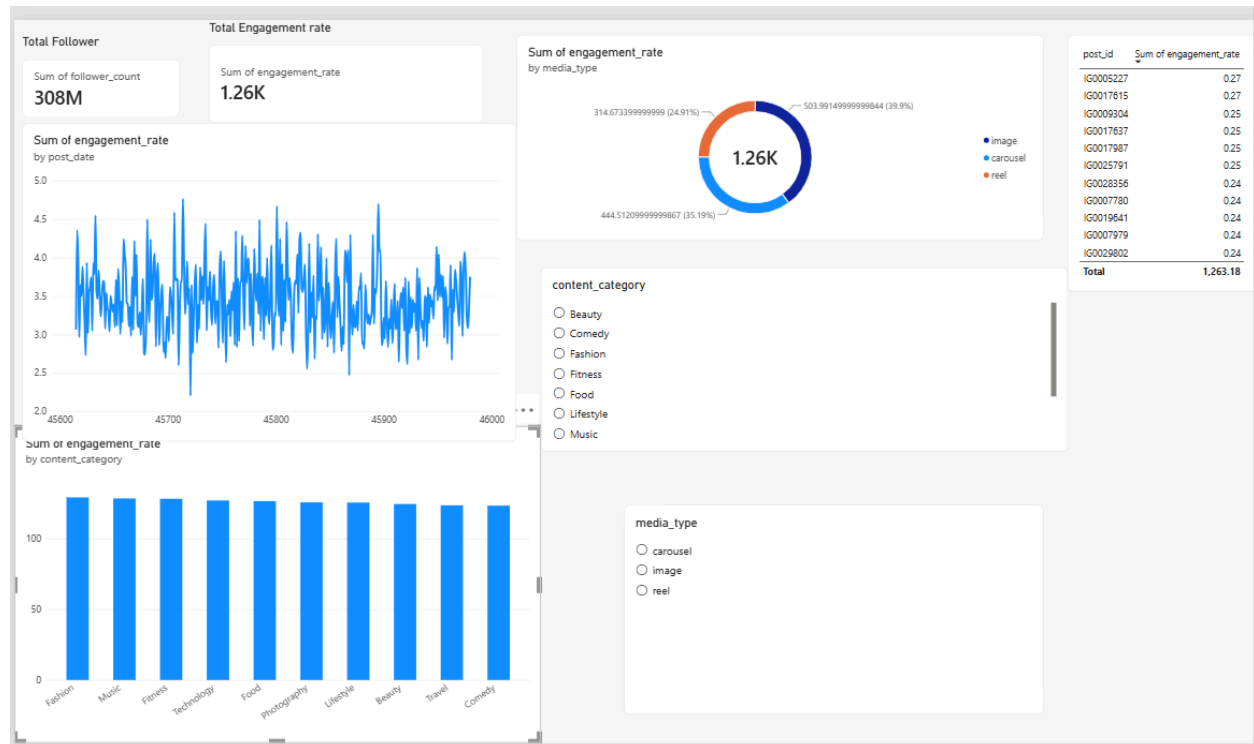
FINAL DASHBOARD

Procedure

1. Arrange the Total Followers and Total Engagement Rate KPI cards at the top.
2. Place the Engagement Over Time line chart in the main analysis area.
3. Place the Engagement by Content Category column chart beside or below it.
4. Place the Engagement by Media Type donut chart in the dashboard.
5. Add the Top Posts by Engagement table.
6. Place the Content Category and Media Type slicers in an easily accessible area.
7. Add the dashboard title:

INSTAGRAM ANALYTICS DASHBOARD

8. Resize and align all visualizations properly.
9. Apply appropriate formatting and titles.
10. Test the slicers to verify that the dashboard responds dynamically.



RESULT :